MACHINE LEARNING PROJECT

Telco Customer Churn Prediction

From exploratory analysis to a deployed Streamlit app

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The Problem and the Data

7,043

customers in the Telco dataset

21

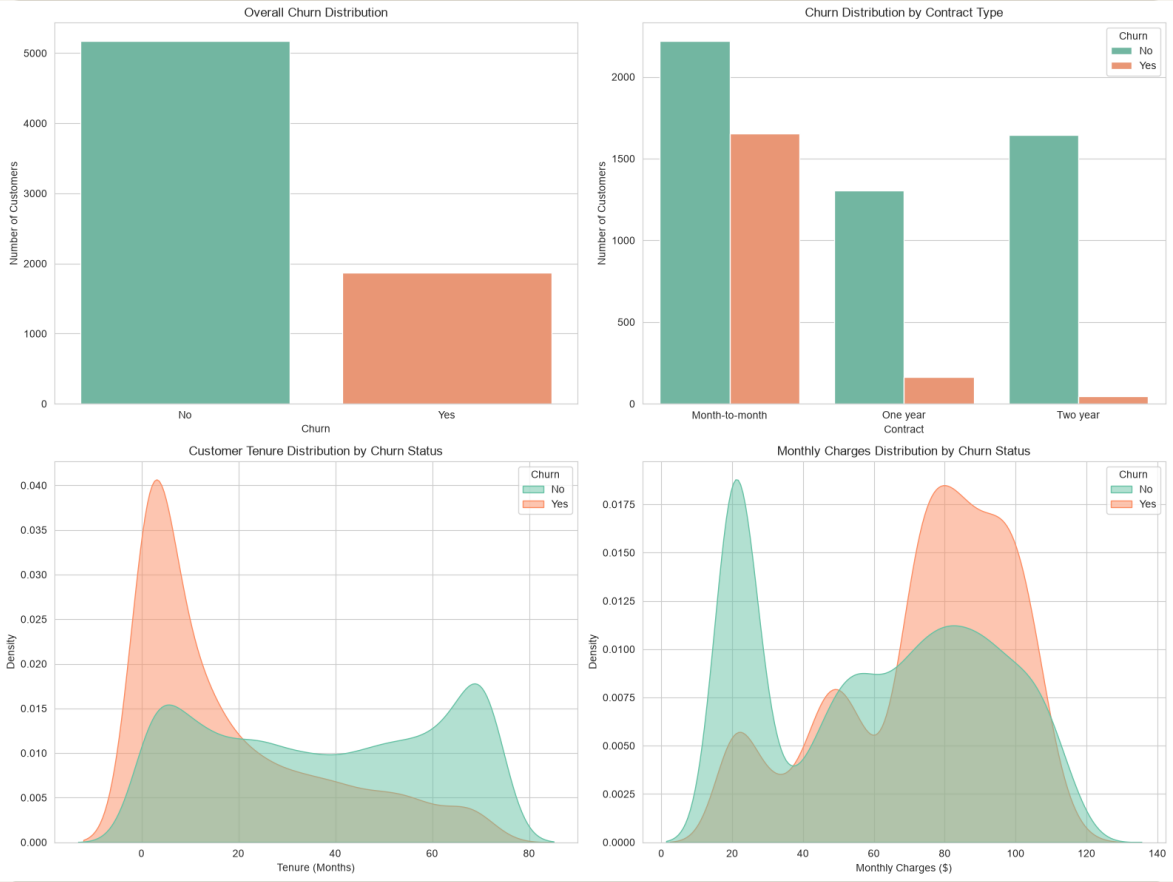
columns: demographics, services, billing

26.5%

churned: 1,869 of 7,043

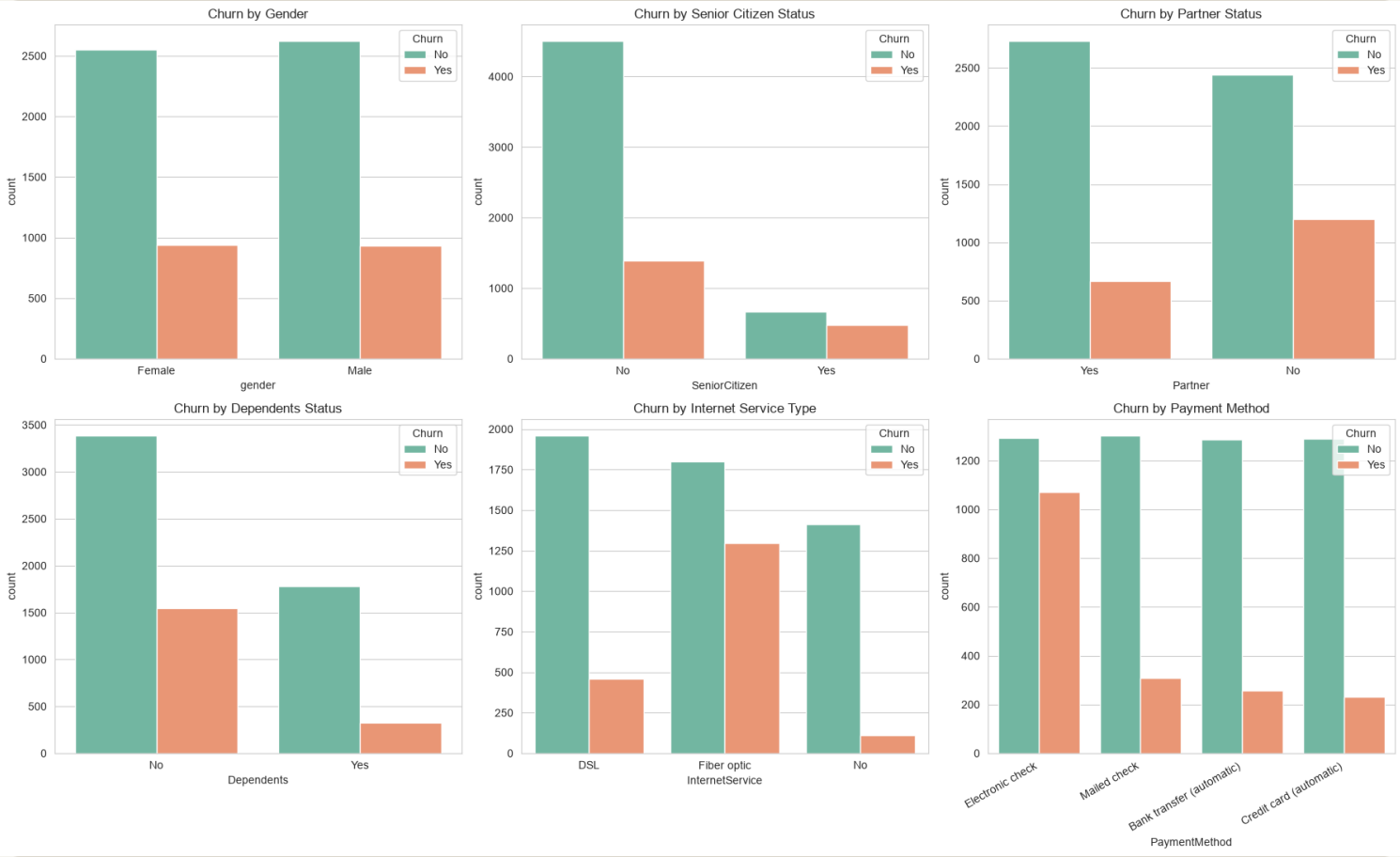
Goal: predict which customers will leave, so retention offers reach them first.

EDA: Churn, Contract and Spend



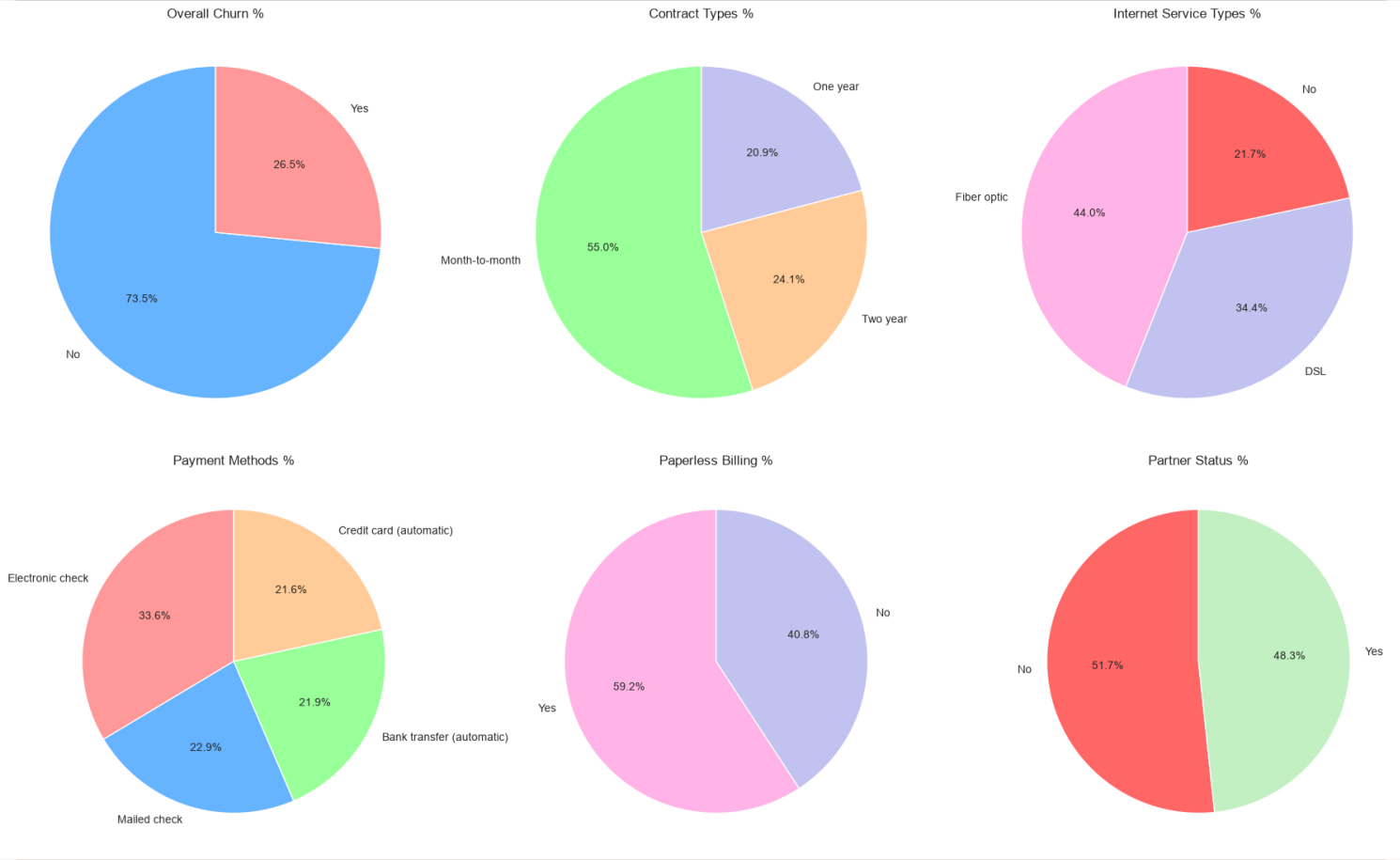
- 26.5% of customers churned
- Month-to-month contracts drive most churn
- New customers (0 to 10 months) leave first
- Churners cluster at $70 to $100 monthly

EDA: Customer Profile and Services



- Gender has almost no effect
- Seniors and customers without partner or dependents churn more
- Fiber optic and electronic check show the highest churn

EDA: Customer Mix



- Share of customers by contract, internet service, payment method, paperless billing and partner status
- Context for the churn rates on the previous slides

What the EDA Shows

Contract

Month-to-month customers churn most; one and two year contracts retain well.

Tenure

Churn concentrates in the first 0 to 10 months, then falls steadily.

Pricing

Churners cluster at $70 to $100 monthly; loyal customers near $20.

Services

Fiber optic internet shows exceptionally high churn.

Payment

Electronic check users churn far more than automatic payers.

Profile

No partner or dependents, and seniors, churn more. Gender does not matter.

Preprocessing and Feature Engineering

80/20
stratified train/test split, seed 42

3
engineered features added

5,323
columns after one-hot encoding

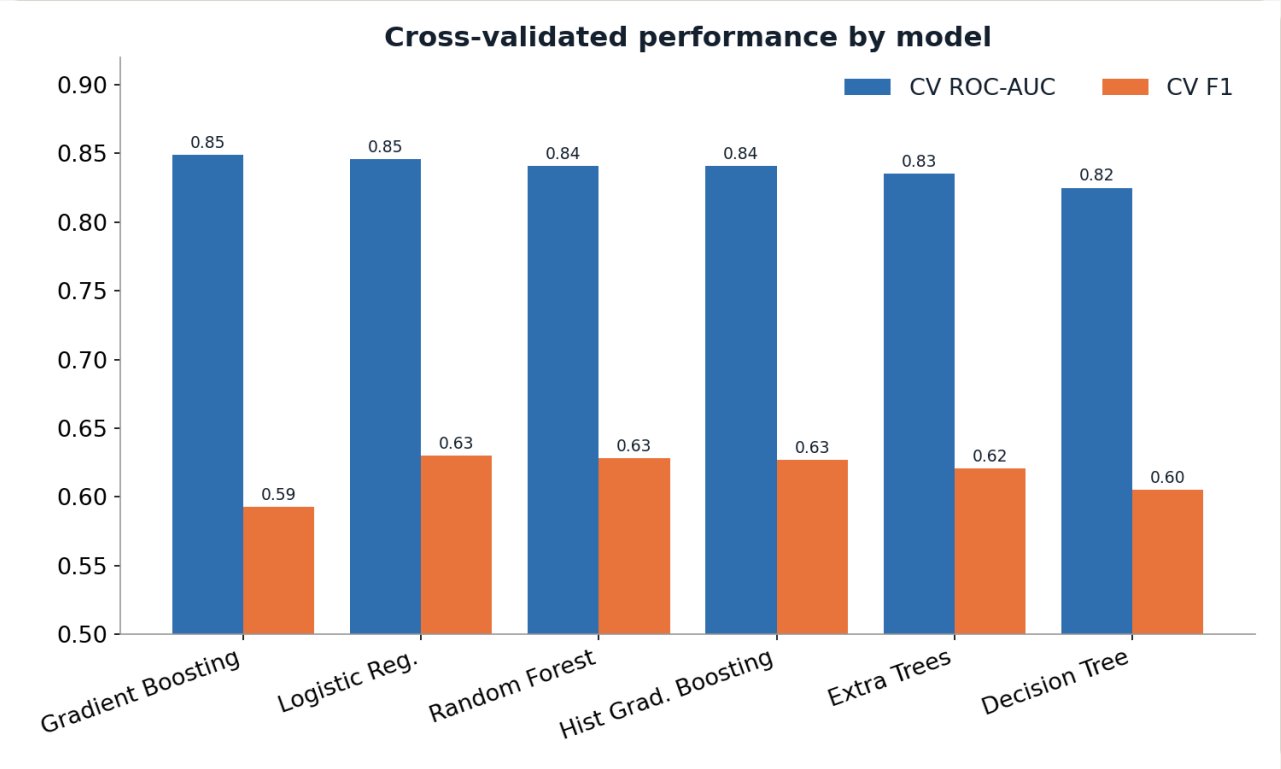
- tenure_monthly_charge_ratio, high_spend_customer, long_term_customer (12+ months)
- Median imputation and scaling for numbers; most-frequent imputation and one-hot for categories
- The high-spend median is learned in fit() on training data only, which avoids leakage

Model Comparison

Model	CV ROC-AUC	CV F1
Gradient Boosting	**0.849**	**0.593**
Logistic Regression	0.846	0.630
Random Forest	0.841	0.628
Hist Gradient Boosting	0.841	0.627
Extra Trees	0.835	0.621
Decision Tree	0.825	0.605

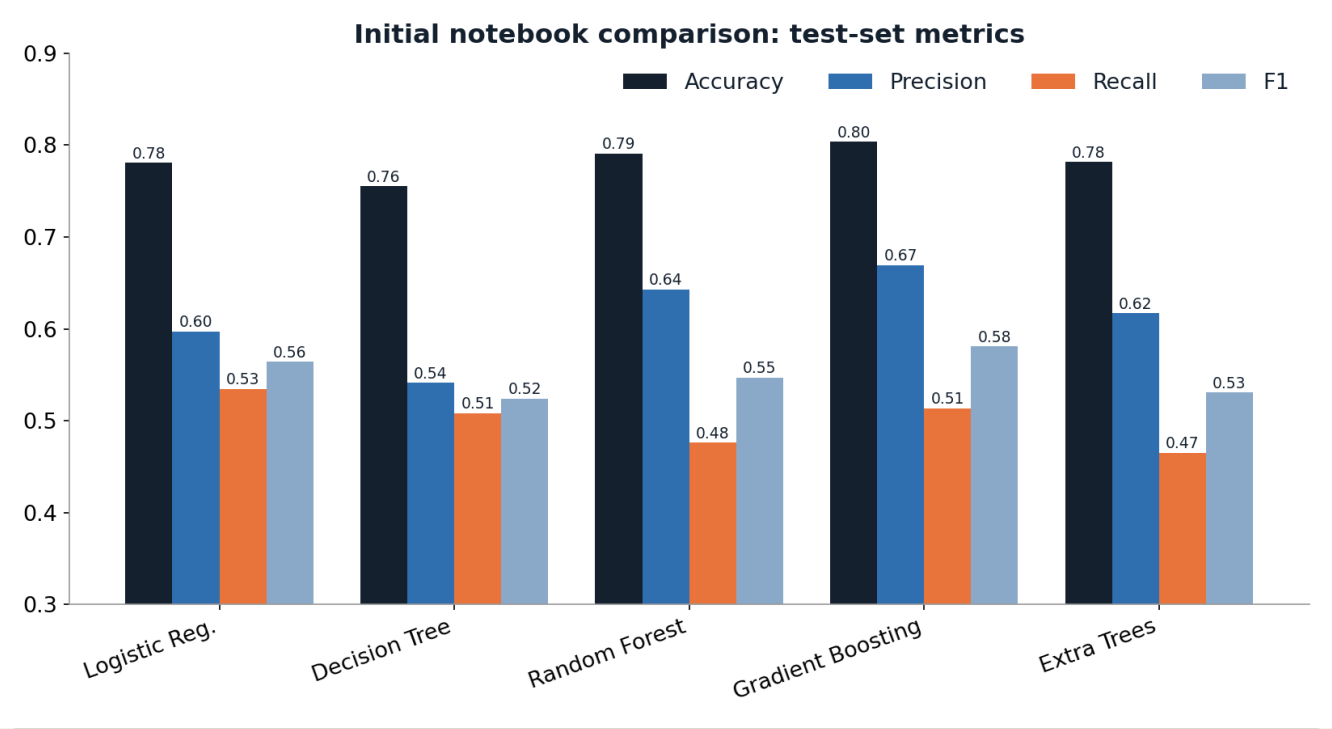
Six models, 5-fold stratified CV, full pipeline inside each fold. Gradient Boosting wins on ROC-AUC; tuning lifted it to 0.851.

Model Comparison Chart



- ROC-AUC is close across models (0.825 to 0.849)
- Gradient Boosting has the best ROC-AUC
- Logistic Regression has the best CV F1 (0.630)

First Pass: Test-Set Metrics



- Notebook run at the default 0.50 threshold
- Gradient Boosting leads on accuracy, precision and F1
- Recall stays near 0.5 for every model, so many churners are missed

Final Model: Gradient Boosting

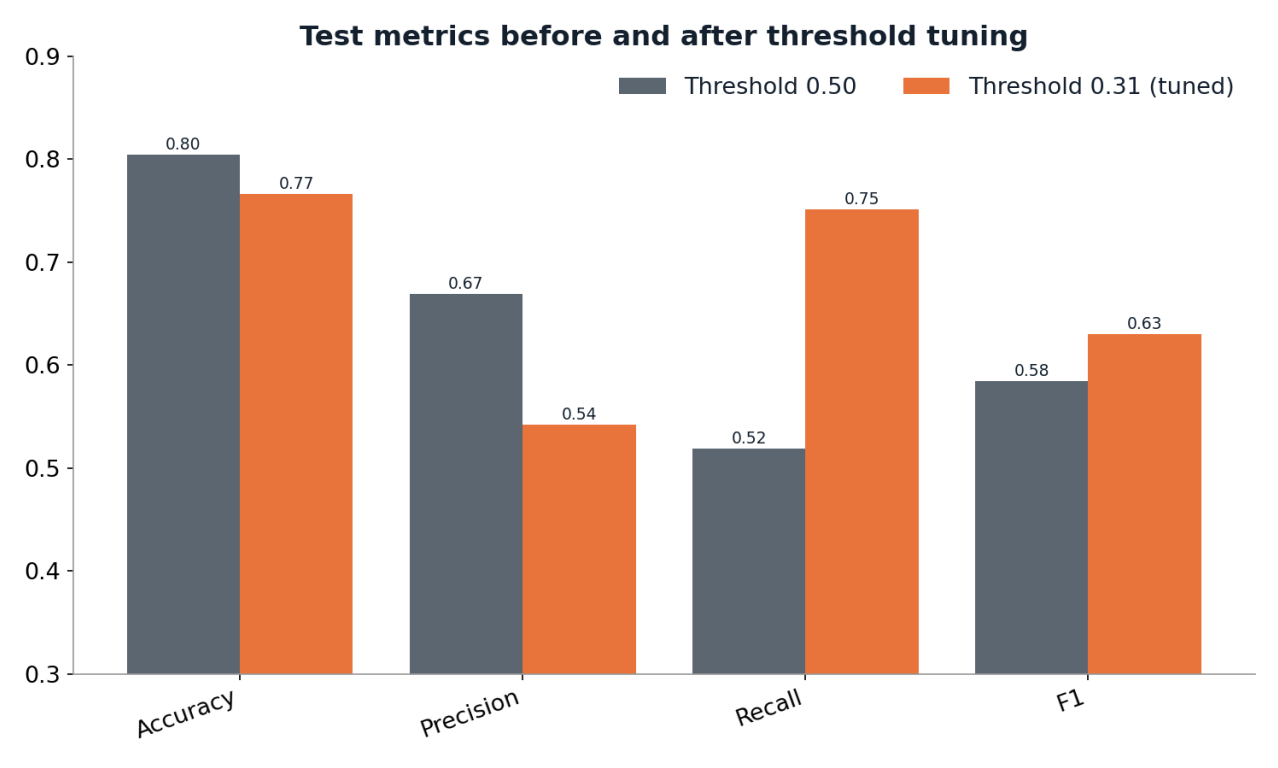
0.845
test ROC-AUC

75%
recall at tuned threshold

0.306
decision threshold (not 0.50)

Moving the threshold from 0.50 to 0.31 raises recall from 52% to 75% and F1 from 0.58 to 0.63. Precision falls from 67% to 54%: more false alarms, far fewer missed churners.

Effect of Threshold Tuning



- Recall rises from 0.52 to 0.75
- F1 rises from 0.58 to 0.63
- Accuracy and precision drop: the cost of catching more churners

Deployment and Next Steps

Streamlit app

Single-customer form with risk label, plus batch CSV scoring and download.

Reproducible

features.py is shared by train.py and app.py; requirements.txt pins every version.

Next

Deploy to Streamlit Cloud, add SHAP explanations, track churn drift over time.